Genetic-Algorithm-Based Parallelism Design Space Exploration for Transformer Accelerators in FINN/Brainsmith

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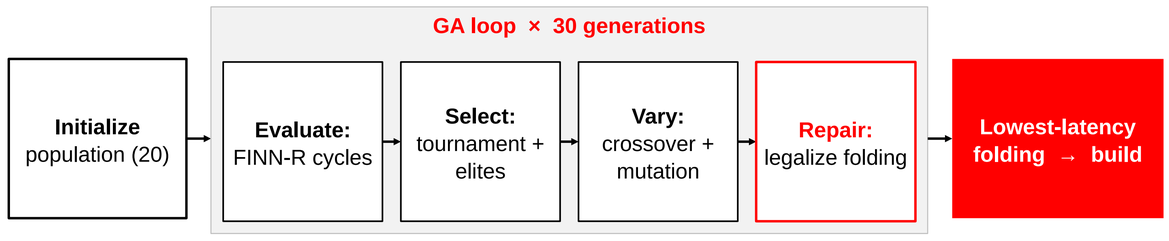
Motivation

- FINN builds dataflow accelerators for quantized neural networks. Each node's **PE** and **SIMD** folding factors set its speed and hardware cost.
- The built-in **SetFolding** heuristic is greedy: given a target FPS, it raises each node's PE and SIMD until that node meets its cycle budget.
- Tuning nodes in isolation ignores dependencies along the **critical path** and leaves substantial latency unexploited.
- Prior work already used GA-based folding DSE in FINN — but only for CNNs, not the transformers behind today's LLMs.
- **FINN-T** and **Brainsmith** add transformer support, but still rely on the greedy heuristic or manual folding.

Goal: tune all folding factors together, automatically.

Key ideas

- **Global folding DSE** in the FINN/Brainsmith flow: a genetic algorithm tunes the PE and SIMD of all nodes together.
- **Genotype:** one integer per tunable parameter, indexing that node's list of valid PE or SIMD values.
- **Fitness:** estimated critical-path cycle count from the FINN-R cost model — a whole-model objective.
- **Repair:** aligns producer–consumer folding; keeps DWC (DataWidthConverter) + FIFO settings legal, else reverts to baseline.
- **Drop-in:** a Brainsmith build step, applicable in any blueprint.



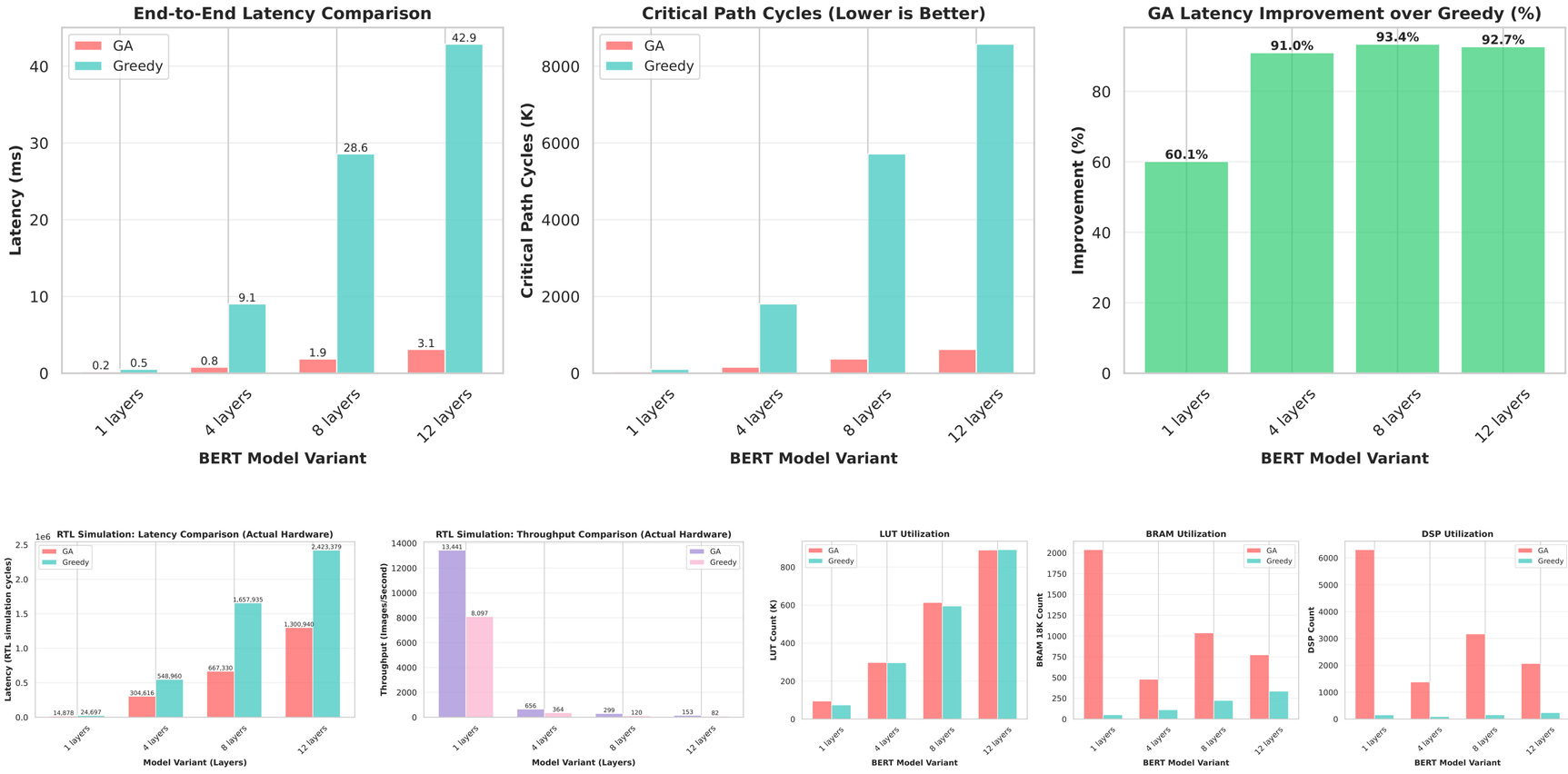
Contribution

- **First for transformers:** global GA folding DSE in FINN/Brainsmith; prior FINN GA work covered only CNNs.
- **Search that scales:** whole-graph search reaches foldings the node-by-node greedy heuristic never finds.
- **Practical:** a drop-in Brainsmith build step; every candidate is buildable, gains confirmed in RTL on Alveo V80.

−50.9%
average latency reduction

+67.9%
higher average throughput

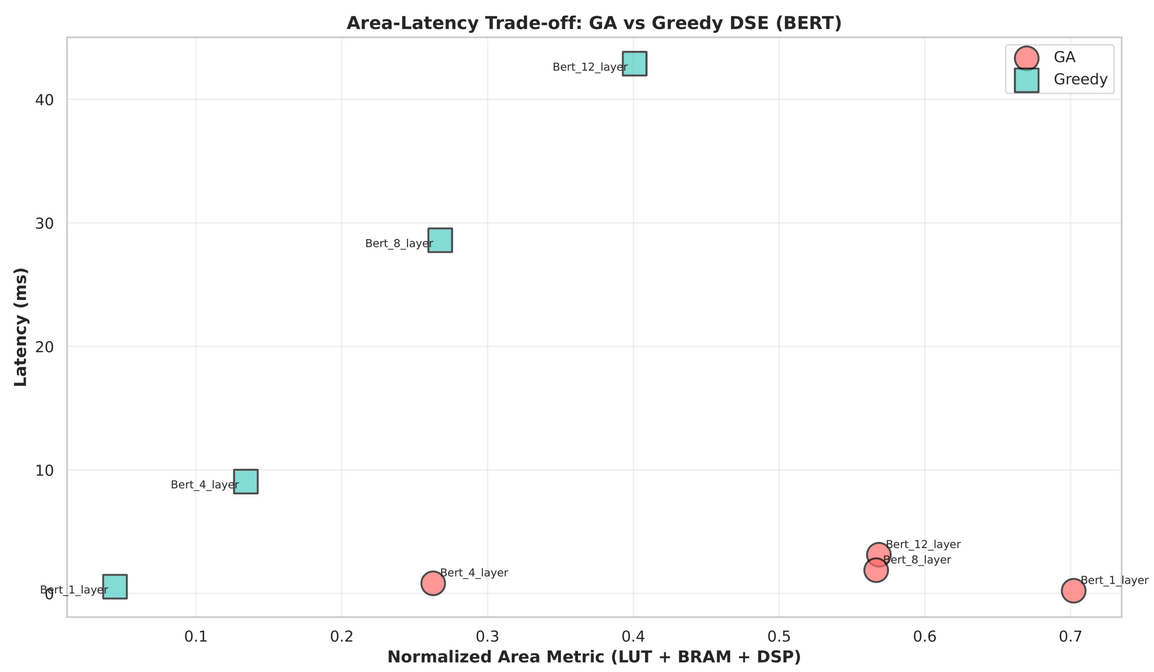
GA beats greedy at every model size — up to 59.7% lower latency, up to 2.5× the throughput



Setup: BERT encoders (1–12 layers) on AMD Alveo V80; GA and greedy run at the same throughput target; measured in RTL simulation.

Discussion

- The GA concentrates SIMD on the dominant **MVAU** (matrix–vector unit) stages (mean ≈ 30–45, up to 256) while PE stays near 1; greedy stops at SIMD 1–2 — the 8-layer critical path drops from 5.7 M to 0.38 M cycles.
- **Area–latency trade-off:** the GA buys latency with DSP and BRAM at the MVAU (unrolling, wider ports, buffering); LUT growth stays modest. The figure at right shows the overall area–latency trade-off.



Area–latency trade-off. *Each point is one BERT variant (1–12 layers); area = normalized LUT + BRAM + DSP. Greedy (teal squares) stays at low area but high latency; GA (red circles) spends area to cut latency by up to 15×.*

Future Work

- Tune the GA hyperparameters — the untuned defaults (population 20, 30 generations) already beat the greedy baseline.
- Make fitness resource-aware: trade latency against DSP, BRAM and LUT budgets (multi-objective search).
- Scale to full-size BERT and other blueprints in Brainsmith.
- Validate designs on the board, beyond RTL simulation.

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